

Timeline (≈ 40 sec)

The project will be completed from January to November.

At the start, we define the problem and set the research goals.

Then, we study related research papers and identify the research gap.

After that, we collect and prepare the Bangla and Banglish datasets.

Next, we design the research method and choose suitable transformer models.

Then, we train and evaluate the models and also analyze hypothesis-only bias using explainable AI.

After that, we compare results and analyze model behavior.

Finally, we write the thesis and prepare the final report and presentation.

The Gantt chart helps us organize all tasks and track progress smoothly.

Time e na kulai le

The project runs from January to November.

First, we define the problem and review related work.

Then, we prepare the dataset and design the method.

After that, we select and train transformer models.

Next, we evaluate the models and analyze hypothesis-only bias.

Finally, we compare results and complete the thesis writing.

The Gantt chart helps us organize all tasks and track progress smoothly.

Budget ($\approx 25\text{--}30$ sec)

The total budget is 71,000 BDT.

It includes costs for GPU, internet, storage, and printing.

Most of the software tools are free.

So, this budget supports the overall research work and implementation.

CEP mapping

Our thesis follows the CEP requirements of knowledge, problem, and activity attributes.

For the knowledge areas K2 to K8, we used evaluation metrics such as Accuracy, Precision, Recall, F1-score, Confusion Matrix, bias score, and SHAP values. We built a complete NLP pipeline using transformer-based models. We created and used the dataset for training and evaluation. We studied hypothesis-only bias and used SHAP to explain model predictions. The implementation was done using Python, PyTorch, Hugging Face, scikit-learn, and Google Colab. We also reviewed recent research papers on NLI, transformer models, and explainable AI.

For the problem attributes P1 to P7, this research requires knowledge of NLP and transformer models. We worked with both Bangla and Banglish inputs and analyzed model performance and bias. We addressed a less explored research problem in this area. All stages are interconnected, starting from dataset preparation to model evaluation and SHAP-based analysis. We also considered feedback from relevant stakeholders during the design process.

For the activity attributes A1 to A5, we create and used datasets and modern machine learning tools, evaluated multiple transformer-based models, and proposed a Bangla–Banglish NLI framework with SHAP-based explainability. The goal of this work is to better understand model behavior and contribute to make NLI systems more reliable and interpretable.